

RIPHAH INTERNATIONAL UNIVERSITY



Software Engineering

Project Title: Campus Hub

Batch: Fall 2023

Project Team Members

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Implementation Phase :

➤ **Create Student Record CRUD Functionality**

- o Implement a form for admins to input new student records.
- o Add functionality for viewing, editing, and deleting student records.
- o Status : Completed.

1. Display Student Data :

```
package main;

import java.io.BufferedReader;
import java.io.File;
import java.io.FileReader;
import java.io.IOException;
import static java.lang.System.exit;
import javax.swing.JOptionPane;

public class DisplayStudentData{
    public String readStudentInfo() {
        File file = new File("Studentdata.txt");
        StringBuilder sb = new StringBuilder();
        try {
            FileReader reader = new FileReader(file);
            BufferedReader breader = new BufferedReader(reader);
            try
            {
                String line;
                while ((line = breader.readLine()) != null) {
                    sb.append(line).append("\n");
                }
            }
            finally
            {
                breader.close();
                reader.close();
            }
        } catch (IOException e) {
            e.printStackTrace();
            JOptionPane.showMessageDialog(null, "Error occurred while reading student information.", "Error",
JOptionPane.ERROR_MESSAGE);
        }
        return sb.toString();
    }
    public String readStudentCourses() {
        File file = new File("EnrolledCourses.txt");
        StringBuilder sb = new StringBuilder();
        try {
            FileReader reader = new FileReader(file);
            BufferedReader breader = new BufferedReader(reader);
            String line;
```

```

        while ((line = breader.readLine()) != null) {
            sb.append(line).append("\n");
        }
        breader.close();
        reader.close();
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while reading student information.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
    return sb.toString();
}
}
class DisplayStudentDataGUI extends javax.swing.JFrame {

    public DisplayStudentDataGUI() {
        initComponents();
    }

    // <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents
    private void initComponents() {

        jPanel1 = new javax.swing.JPanel();
        logoutButton = new javax.swing.JButton();
        backButton = new javax.swing.JButton();
        PowerOffButton = new javax.swing.JButton();
        jLabel1 = new javax.swing.JLabel();
        courseButton = new javax.swing.JButton();
        displayStudentInfo = new javax.swing.JButton();
        jScrollPane1 = new javax.swing.JScrollPane();
        jTextArea1 = new javax.swing.JTextArea();

        setDefaultCloseOperation(javax.swing.WindowConstants.DISPOSE_ON_CLOSE);

        jPanel1.setBackground(new java.awt.Color(255, 210, 255));

        logoutButton.setBackground(new java.awt.Color(255, 204, 255));
        logoutButton.setFont(new java.awt.Font("Lucida Sans Typewriter", 0, 14)); // NOI18N
        logoutButton.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/logout.png"))); // NOI18N
        logoutButton.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                logoutButtonActionPerformed(evt);
            }
        });

        backButton.setBackground(new java.awt.Color(255, 204, 255));
        backButton.setFont(new java.awt.Font("Lucida Sans Typewriter", 0, 14)); // NOI18N
        backButton.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/back.png"))); // NOI18N
        backButton.addActionListener(new java.awt.event.ActionListener() {
            public void actionPerformed(java.awt.event.ActionEvent evt) {
                backButtonActionPerformed(evt);
            }
        });

        PowerOffButton.setBackground(new java.awt.Color(255, 204, 255));
        PowerOffButton.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/off.png"))); // NOI18N

```

```
PowerOffButton.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        PowerOffButton.ActionPerformed(evt);
    }
});
```

```
jLabel1.setFont(new java.awt.Font("Segoe Print", 0, 18)); // NOI18N
jLabel1.setText("Display Student Info");
```

```
courseButton.setBackground(new java.awt.Color(255, 204, 255));
courseButton.setText("Display Enrolled Courses");
courseButton.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        courseButtonActionPerformed(evt);
    }
});
```

```
displayStudentInfo.setBackground(new java.awt.Color(255, 204, 255));
displayStudentInfo.setText("Display Student Info");
displayStudentInfo.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        displayStudentInfoActionPerformed(evt);
    }
});
```

```
jTextArea1.setBackground(new java.awt.Color(255, 204, 255));
jTextArea1.setColumns(20);
jTextArea1.setRows(5);
jScrollPane1.setViewportView(jTextArea1);
```

```

    javax.swing.GroupLayout jPanel1Layout = new javax.swing.GroupLayout(jPanel1);
    jPanel1.setLayout(jPanel1Layout);
    jPanel1Layout.setHorizontalGroup(
        jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(javax.swing.GroupLayout.Alignment.TRAILING, jPanel1Layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(PowerOffButton, javax.swing.GroupLayout.PREFERRED_SIZE, 86,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(backButton)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(logoutButton, javax.swing.GroupLayout.PREFERRED_SIZE, 77,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap(javax.swing.GroupLayout.DEFAULT_SIZE, true)
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 442,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap()
                .addComponent(jLabel1)
                .addContainerGap())
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addComponent(displayStudentInfo)
                .addContainerGap())
    );
    jPanel1Layout.setVerticalGroup(
        jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addGroup(jPanel1Layout.createSequentialGroup()
                .addContainerGap()
                .addComponent(PowerOffButton, javax.swing.GroupLayout.PREFERRED_SIZE, 86,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(backButton, javax.swing.GroupLayout.PREFERRED_SIZE, 40,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(logoutButton, javax.swing.GroupLayout.PREFERRED_SIZE, 40,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addPreferredGap(javax.swing.LayoutStyle.ComponentPlacement.RELATED)
                .addComponent(jScrollPane1, javax.swing.GroupLayout.PREFERRED_SIZE, 300,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap()
                .addComponent(jLabel1, javax.swing.GroupLayout.PREFERRED_SIZE, 40,
                    javax.swing.GroupLayout.PREFERRED_SIZE)
                .addContainerGap())
    );
}


```

```

        .addComponent(courseButton))))
        .addContainerGap(28, Short.MAX_VALUE))))
    );
    jPanel1Layout.setVerticalGroup(
        jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(jPanel1Layout.createSequentialGroup())
        .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(PowerOffButton, javax.swing.GroupLayout.PREFERRED_SIZE, 82,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(jLabel1, javax.swing.GroupLayout.Alignment.TRAILING))
        .addGap(18, 18, 18)
        .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.BASELINE)
            .addComponent(displayStudentInfo, javax.swing.GroupLayout.PREFERRED_SIZE, 52,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(courseButton, javax.swing.GroupLayout.PREFERRED_SIZE, 52,
                javax.swing.GroupLayout.PREFERRED_SIZE))
        .addGap(18, 18, 18)
        .addComponent(jScrollPane1, javax.swing.GroupLayout.DEFAULT_SIZE, 164, Short.MAX_VALUE)
        .addGap(65, 65, 65)
        .addGroup(jPanel1Layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
            .addComponent(loginButton, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
                javax.swing.GroupLayout.PREFERRED_SIZE)
            .addComponent(backButton, javax.swing.GroupLayout.PREFERRED_SIZE, 71,
                javax.swing.GroupLayout.PREFERRED_SIZE))
        .addContainerGap())
    );

    javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
    getContentPane().setLayout(layout);
    layout.setHorizontalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
            javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
    );
    layout.setVerticalGroup(
        layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
            javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
    );

    pack();
    setLocationRelativeTo(null);
} // </editor-fold> // GEN-END: initComponents

```

```

private void logoutButtonActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_logoutButtonActionPerformed
    this.toBack();
    LoginPage login = new LoginPage();
    login.setVisible(true);
    login.toFront();
} // GEN-LAST:event_logoutButtonActionPerformed

```

```

private void backButtonActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_backButtonActionPerformed
    this.toBack();
}

```

```

    }//GEN-LAST:event_backButtonActionPerformed

    private void PowerOffButtonActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_PowerOffButtonActionPerformed
        exit(0);
    }//GEN-LAST:event_PowerOffButtonActionPerformed

    private void displayStudentInfoActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_displayStudentInfoActionPerformed
        DisplayStudentData displayStudentData = new DisplayStudentData();
        String studentInfo = displayStudentData.readStudentInfo();
        jTextArea1.setText(studentInfo);
    }//GEN-LAST:event_displayStudentInfoActionPerformed

    private void courseButtonActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_courseButtonActionPerformed
        DisplayStudentData displayStudentData = new DisplayStudentData();
        String studentInfo = displayStudentData.readStudentCourses();
        jTextArea1.setText(studentInfo);
    }//GEN-LAST:event_courseButtonActionPerformed

    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JButton PowerOffButton;
    private javax.swing.JButton backButton;
    private javax.swing.JButton courseButton;
    private javax.swing.JButton displayStudentInfo;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JPanel jPanel1;
    private javax.swing.JScrollPane jScrollPane1;
    private javax.swing.JTextArea jTextArea1;
    private javax.swing.JButton logoutButton;
    // End of variables declaration//GEN-END:variables
}

```

2. Student Information :

```

package main;

import java.io.BufferedWriter;
import java.io.File;
import java.io.*;
import java.io.IOException;
import static java.lang.System.exit;
import javax.swing.JOptionPane;
import java.util.ArrayList;

public class Studentinfo {

    //data members
    private String name;
    private String id;
    private String age;
    private String address;
    private String personalContactNumber;
}

```

```
private String emergencyContactNumber;
private String gender;
private String nationality;
private String dob;
private String email;

//Constructor
public Studentinfo()
{
    name = null;
    id = null;
    age = null;
    address = null;
    personalContactNumber = null;
    emergencyContactNumber = null;
    gender = null;
    nationality = null;
    dob = null;
    email = null;
}

public Studentinfo(String name, String id, String age, String address, String personalContactNumber, String
emergencyContactNumber, String gender, String nationality, String dob, String email)
{
    this.name = name;
    this.id = id;
    this.age = age;
    this.address = address;
    this.personalContactNumber = personalContactNumber;
    this.emergencyContactNumber = emergencyContactNumber;
    this.gender = gender;
    this.nationality = nationality;
    this.dob = dob;
    this.email = email;
}

// Getters
public String getName() {
    return name;
}

public String getId() {
    return id;
}

public String getAge() {
    return age;
}

public String getAddress() {
    return address;
}

public String getPersonalContactNumber() {
    return personalContactNumber;
}
```

```
}

public String getEmergencyContactNumber() {
    return emergencyContactNumber;
}

public String getGender() {
    return gender;
}

public String getNationality() {
    return nationality;
}

public String getDob() {
    return dob;
}

public String getEmail() {
    return email;
}

// Setters
public void setName(String name) {
    this.name = name;
}

public void setId(String id) {
    this.id = id;
}

public void setAge(String age) {
    this.age = age;
}

public void setAddress(String address) {
    this.address = address;
}

public void setPersonalContactNumber(String personalContactNumber) {
    this.personalContactNumber = personalContactNumber;
}

public void setEmergencyContactNumber(String emergencyContactNumber) {
    this.emergencyContactNumber = emergencyContactNumber;
}

public void setGender(String gender) {
    this.gender = gender;
}

public void setNationality(String nationality) {
    this.nationality = nationality;
}

public void setDob(String dob) {
```



```

        this.dob = dob;
    }

    public void setEmail(String email) {
        this.email = email;
    }

    // Display info
    @Override
    public String toString() {
        return "Student Information\nName: " + name + "\n" +
            "ID: " + id + "\n" +
            "Age: " + age + "\n" +
            "Address: " + address + "\n" +
            "Personal Contact Number: " + personalContactNumber + "\n" +
            "Emergency Contact Number: " + emergencyContactNumber + "\n" +
            "Gender: " + gender + "\n" +
            "Nationality: " + nationality + "\n" +
            "Date of Birth: " + dob + "\n" +
            "Email: " + email + "\n";
    }
}

class StudentinfoGUI extends javax.swing.JFrame {

    /**
     * Creates new form StudentInfo
     */
    public StudentinfoGUI() {
        initComponents();
    }

    private void saveStudentInfo() {
        File file = new File("Studentdata.txt");
        try {
            FileWriter writer = new FileWriter(file);
            BufferedWriter bwriter = new BufferedWriter(writer);
            try {
                String name = nameField.getText().trim();
                String id = idField.getText().trim();
                String age = ageField.getText().trim();
                String address = addressField.getText().trim();
                String personalContactNumber = personalContactNumberField.getText().trim();
                String emergencyContactNumber = emergencyContactNumberField.getText().trim();
                String gender = genderField.getText().trim();
                String nationality = nationalityField.getText().trim();
                String dob = dobField.getText().trim();
                String email = emailField.getText().trim();

                // Validate inputs
                if (name.isEmpty() || id.isEmpty() || age.isEmpty() || address.isEmpty() ||
                    personalContactNumber.isEmpty() || emergencyContactNumber.isEmpty() ||
                    gender.isEmpty() || nationality.isEmpty() || dob.isEmpty() || email.isEmpty()) {
                    throw new IllegalArgumentException("All fields must be filled.");
                }
            }
        }
    }
}

```

```

int ageInt;
try {
    ageInt = Integer.parseInt(age);
    if (ageInt <= 0) {
        throw new IllegalArgumentException("Age must be a positive number.");
    }
} catch (NumberFormatException e) {
    throw new IllegalArgumentException("Age must be a valid number.", e);
}

if (!personalContactNumber.matches("\\d+")) {
    throw new IllegalArgumentException("Personal Contact Number must contain only digits.");
}

if (!emergencyContactNumber.matches("\\d+")) {
    throw new IllegalArgumentException("Emergency Contact Number must contain only digits.");
}

if (!email.matches("^\\w.%+-]+@[\\w.-]+\\.[a-zA-Z]{2,6}$")) {
    throw new IllegalArgumentException("Email is not in a valid format.");
}

Studentinfo student = new Studentinfo(name, id, age, address, personalContactNumber,
emergencyContactNumber, gender, nationality, dob, email);
bwriter.write(student.toString());

JOptionPane.showMessageDialog(this, "Student Information saved to Studentdata.txt.");
} catch (IllegalArgumentException e) {
    JOptionPane.showMessageDialog(this, e.getMessage(), "Input Error", JOptionPane.ERROR_MESSAGE);
} finally {
    try {
        bwriter.close();
        writer.close();
    } catch (IOException e) {
        e.printStackTrace();
    }
}
} catch (IOException e) {
    e.printStackTrace();
    JOptionPane.showMessageDialog(this, "Error occurred while saving student information.", "Error",
JOptionPane.ERROR_MESSAGE);
}

}

```

```

@SuppressWarnings("unchecked")

```

```

// <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents

```

```

private void initComponents() {

```

```

    jButton1 = new javax.swing.JButton();
    jPanel1 = new javax.swing.JPanel();
    jLabel2 = new javax.swing.JLabel();
    jLabel3 = new javax.swing.JLabel();
    jLabel5 = new javax.swing.JLabel();

```

```

jLabel6 = new javax.swing.JLabel();
jLabel8 = new javax.swing.JLabel();
jLabel9 = new javax.swing.JLabel();
jLabel10 = new javax.swing.JLabel();
jLabel11 = new javax.swing.JLabel();
jLabel4 = new javax.swing.JLabel();
jLabel7 = new javax.swing.JLabel();
jLabel1 = new javax.swing.JLabel();
saveButton = new javax.swing.JButton();
nameField = new javax.swing.JTextField();
idField = new javax.swing.JTextField();
ageField = new javax.swing.JTextField();
emailField = new javax.swing.JTextField();
nationalityField = new javax.swing.JTextField();
addressField = new javax.swing.JTextField();
personalContactNumberField = new javax.swing.JTextField();
dobField = new javax.swing.JTextField();
emergencyContactNumberField = new javax.swing.JTextField();
genderField = new javax.swing.JTextField();
genderComboBox = new javax.swing.JComboBox<>();
clearButton = new javax.swing.JButton();
logoutButton = new javax.swing.JButton();
jButton11 = new javax.swing.JButton();
jButton12 = new javax.swing.JButton();

jButton1.setText("jButton1");

setDefaultCloseOperation(javax.swing.WindowConstants.DISPOSE_ON_CLOSE);

jPanel1.setBackground(new java.awt.Color(255, 210, 255));
jPanel1.setLayout(new org.netbeans.lib.awtextra.AbsoluteLayout());

jLabel2.setBackground(new java.awt.Color(255, 204, 255));
jLabel2.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel2.setText("Email");
jPanel1.add(jLabel2, new org.netbeans.lib.awtextra.AbsoluteConstraints(37, 358, 80, 30));

jLabel3.setBackground(new java.awt.Color(255, 204, 255));
jLabel3.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel3.setText("Nationality");
jPanel1.add(jLabel3, new org.netbeans.lib.awtextra.AbsoluteConstraints(339, 124, 80, 30));

jLabel5.setBackground(new java.awt.Color(255, 204, 255));
jLabel5.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel5.setText("Date Of Birth");
jLabel5.setMaximumSize(new java.awt.Dimension(80, 30));
jPanel1.add(jLabel5, new org.netbeans.lib.awtextra.AbsoluteConstraints(339, 242, 80, 30));

jLabel6.setBackground(new java.awt.Color(255, 204, 255));
jLabel6.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel6.setText("Address");
jPanel1.add(jLabel6, new org.netbeans.lib.awtextra.AbsoluteConstraints(339, 180, 80, 30));

jLabel8.setBackground(new java.awt.Color(255, 204, 255));
jLabel8.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel8.setText("Age");

```

```

jPanel1.add(jLabel8, new org.netbeans.lib.awtextra.AbsoluteConstraints(31, 242, 80, 30));

jLabel9.setBackground(new java.awt.Color(255, 204, 255));
jLabel9.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel9.setText("Emergency Contact Number");
jPanel1.add(jLabel9, new org.netbeans.lib.awtextra.AbsoluteConstraints(339, 358, 165, 30));

jLabel10.setBackground(new java.awt.Color(255, 204, 255));
jLabel10.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel10.setText("ID");
jPanel1.add(jLabel10, new org.netbeans.lib.awtextra.AbsoluteConstraints(31, 180, 80, 30));

jLabel11.setBackground(new java.awt.Color(255, 204, 255));
jLabel11.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel11.setText("Name");
jPanel1.add(jLabel11, new org.netbeans.lib.awtextra.AbsoluteConstraints(33, 124, 80, 30));

jLabel4.setBackground(new java.awt.Color(255, 204, 255));
jLabel4.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel4.setText("Gender");
jLabel4.setPreferredSize(new java.awt.Dimension(80, 30));
jPanel1.add(jLabel4, new org.netbeans.lib.awtextra.AbsoluteConstraints(31, 296, -1, -1));

jLabel7.setBackground(new java.awt.Color(255, 204, 255));
jLabel7.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
jLabel7.setText("Personal Contact Number");
jPanel1.add(jLabel7, new org.netbeans.lib.awtextra.AbsoluteConstraints(339, 296, 155, 30));

jLabel1.setFont(new java.awt.Font("Segoe Print", 1, 18)); // NOI18N
jLabel1.setHorizontalAlignment(javax.swing.SwingConstants.CENTER);
jLabel1.setText("Enter Your Particulars");
jPanel1.add(jLabel1, new org.netbeans.lib.awtextra.AbsoluteConstraints(230, 40, 275, 34));

saveButton.setBackground(new java.awt.Color(255, 204, 255));
saveButton.setFont(new java.awt.Font("Dialog", 0, 14)); // NOI18N
saveButton.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/save.png"))); // NOI18N
saveButton.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        saveButtonActionPerformed(evt);
    }
});
jPanel1.add(saveButton, new org.netbeans.lib.awtextra.AbsoluteConstraints(240, 430, 90, 90));

nameField.setBackground(new java.awt.Color(255, 204, 255));
nameField.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
nameField.setText("Enter your Name");
nameField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        nameFieldActionPerformed(evt);
    }
});
jPanel1.add(nameField, new org.netbeans.lib.awtextra.AbsoluteConstraints(125, 117, 179, 44));

idField.setBackground(new java.awt.Color(255, 204, 255));
idField.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
idField.setText("Enter your ID");

```

```

idField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        idFieldActionPerformed(evt);
    }
});
jPanel1.add(idField, new org.netbeans.lib.awtextra.AbsoluteConstraints(123, 173, 179, 44));

ageField.setBackground(new java.awt.Color(255, 204, 255));
ageField.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
ageField.setText("Enter your Age");
ageField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        ageFieldActionPerformed(evt);
    }
});
jPanel1.add(ageField, new org.netbeans.lib.awtextra.AbsoluteConstraints(123, 235, 179, 44));

emailField.setBackground(new java.awt.Color(255, 204, 255));
emailField.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
emailField.setText("Enter your Email");
emailField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        emailFieldActionPerformed(evt);
    }
});
jPanel1.add(emailField, new org.netbeans.lib.awtextra.AbsoluteConstraints(123, 351, 179, 44));

nationalityField.setBackground(new java.awt.Color(255, 204, 255));
nationalityField.setText("Enter your Nationality");
nationalityField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        nationalityFieldActionPerformed(evt);
    }
});
jPanel1.add(nationalityField, new org.netbeans.lib.awtextra.AbsoluteConstraints(512, 117, 177, 44));

addressField.setBackground(new java.awt.Color(255, 204, 255));
addressField.setText("Enter your Address");
addressField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        addressFieldActionPerformed(evt);
    }
});
jPanel1.add(addressField, new org.netbeans.lib.awtextra.AbsoluteConstraints(512, 173, 177, 44));

personalContactNumberField.setBackground(new java.awt.Color(255, 204, 255));
personalContactNumberField.setText("Enter your Contact");
personalContactNumberField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        personalContactNumberFieldActionPerformed(evt);
    }
});
jPanel1.add(personalContactNumberField, new org.netbeans.lib.awtextra.AbsoluteConstraints(512, 289, 177,
44));

dobField.setBackground(new java.awt.Color(255, 204, 255));

```

```

dobField.setText("Enter your DOB");
dobField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        dobFieldActionPerformed(evt);
    }
});
jPanel1.add(dobField, new org.netbeans.lib.awtextra.AbsoluteConstraints(512, 235, 177, 44));

emergencyContactNumberField.setBackground(new java.awt.Color(255, 204, 255));
emergencyContactNumberField.setText("Enter your Contact");
emergencyContactNumberField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        emergencyContactNumberFieldActionPerformed(evt);
    }
});
jPanel1.add(emergencyContactNumberField, new org.netbeans.lib.awtextra.AbsoluteConstraints(510, 350, 177, 44));

genderField.setBackground(new java.awt.Color(255, 204, 255));
genderField.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
genderField.setText("Enter your Gender");
genderField.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        genderFieldActionPerformed(evt);
    }
});
jPanel1.add(genderField, new org.netbeans.lib.awtextra.AbsoluteConstraints(123, 289, 89, 44));

genderComboBox.setBackground(new java.awt.Color(255, 204, 255));
genderComboBox.setFont(new java.awt.Font("Lucida Sans", 0, 12)); // NOI18N
genderComboBox.setModel(new javax.swing.DefaultComboBoxModel<>(new String[] { "male", "female" }));
genderComboBox.setCursor(new java.awt.Cursor(java.awt.Cursor.DEFAULT_CURSOR));
genderComboBox.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        genderComboBoxActionPerformed(evt);
    }
});
jPanel1.add(genderComboBox, new org.netbeans.lib.awtextra.AbsoluteConstraints(224, 293, 78, 34));

clearButton.setBackground(new java.awt.Color(255, 204, 255));
clearButton.setFont(new java.awt.Font("Lucida Sans Typewriter", 0, 14)); // NOI18N
clearButton.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/reset.png"))); // NOI18N
clearButton.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        clearButtonActionPerformed(evt);
    }
});
jPanel1.add(clearButton, new org.netbeans.lib.awtextra.AbsoluteConstraints(370, 430, 90, 90));

logoutButton.setBackground(new java.awt.Color(255, 204, 255));
logoutButton.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/logout.png"))); // NOI18N
logoutButton.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        logoutButtonActionPerformed(evt);
    }
});

```

```

jPanel1.add(loginButton, new org.netbeans.lib.awtextra.AbsoluteConstraints(610, 480, 90, 90));

jButton11.setBackground(new java.awt.Color(255, 204, 255));
jButton11.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/off.png"))); // NOI18N
jButton11.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton11ActionPerformed(evt);
    }
});
jPanel1.add(jButton11, new org.netbeans.lib.awtextra.AbsoluteConstraints(600, 0, 90, 90));

jButton12.setBackground(new java.awt.Color(255, 204, 255));
jButton12.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/back.png"))); // NOI18N
jButton12.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton12ActionPerformed(evt);
    }
});
jPanel1.add(jButton12, new org.netbeans.lib.awtextra.AbsoluteConstraints(20, 490, 90, 90));

javax.swing.GroupLayout layout = new javax.swing.GroupLayout(getContentPane());
getContentPane().setLayout(layout);
layout.setHorizontalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
);
layout.setVerticalGroup(
    layout.createParallelGroup(javax.swing.GroupLayout.Alignment.LEADING)
        .addGroup(layout.createSequentialGroup()
            .addComponent(jPanel1, javax.swing.GroupLayout.DEFAULT_SIZE,
javax.swing.GroupLayout.DEFAULT_SIZE, Short.MAX_VALUE)
        );

pack();
 setLocationRelativeTo(null);
} // </editor-fold> // GEN-END: initComponents

private void nameFieldActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_nameFieldActionPerformed
    ArrayList<String> Student = new ArrayList<>();
    Student.add(nameField.getText());
} // GEN-LAST:event_nameFieldActionPerformed

private void idFieldActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_idFieldActionPerformed
    // TODO add your handling code here:
} // GEN-LAST:event_idFieldActionPerformed

private void ageFieldActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_ageFieldActionPerformed
    // TODO add your handling code here:
} // GEN-LAST:event_ageFieldActionPerformed

```

```

    private void emailFieldActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_emailFieldActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_emailFieldActionPerformed

    private void nationalityFieldActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_nationalityFieldActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_nationalityFieldActionPerformed

    private void addressFieldActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_addressFieldActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_addressFieldActionPerformed

    private void personalContactNumberFieldActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_personalContactNumberFieldActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_personalContactNumberFieldActionPerformed

    private void dobFieldActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_dobFieldActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_dobFieldActionPerformed

    private void emergencyContactNumberFieldActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_emergencyContactNumberFieldActionPerformed
        // TODO add your handling code here:
    }//GEN-LAST:event_emergencyContactNumberFieldActionPerformed

    private void genderFieldActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_genderFieldActionPerformed

    }//GEN-LAST:event_genderFieldActionPerformed

    private void saveButtonActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_saveButtonActionPerformed
        saveStudentInfo();
    }//GEN-LAST:event_saveButtonActionPerformed

    private void genderComboBoxActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_genderComboBoxActionPerformed
        genderField.setText(genderComboBox.getSelectedItem().toString());
        genderField.setEditable(false);
    }//GEN-LAST:event_genderComboBoxActionPerformed

    private void clearButtonActionPerformed(java.awt.event.ActionEvent evt) {GEN-
FIRST:event_clearButtonActionPerformed
        nameField.setText("");
        idField.setText("");
        ageField.setText("");
        addressField.setText("");
        personalContactNumberField.setText("");
        emergencyContactNumberField.setText("");
        genderComboBox.setSelectedIndex(0);
        nationalityField.setText("");

```



```

        dobField.setText("");
        emailField.setText("");
    }//GEN-LAST:event_clearButtonActionPerformed

    private void logoutButtonActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_logoutButtonActionPerformed
        this.toBack();
        LoginPage login = new LoginPage();
        login.setVisible(true);
        login.toFront();
    }//GEN-LAST:event_logoutButtonActionPerformed

    private void jButton11ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_jButton11ActionPerformed
        exit(0);
    }//GEN-LAST:event_jButton11ActionPerformed

    private void jButton12ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-FIRST:event_jButton12ActionPerformed
        this.toBack();
        Student student = new Student();
        student.setVisible(true);
        student.toFront();
    }//GEN-LAST:event_jButton12ActionPerformed

    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JTextField addressField;
    private javax.swing.JTextField ageField;
    private javax.swing.JButton clearButton;
    private javax.swing.JTextField dobField;
    private javax.swing.JTextField emailField;
    private javax.swing.JTextField emergencyContactNumberField;
    private javax.swing.JComboBox<String> genderComboBox;
    private javax.swing.JTextField genderField;
    private javax.swing.JTextField idField;
    private javax.swing.JButton jButton1;
    private javax.swing.JButton jButton11;
    private javax.swing.JButton jButton12;
    private javax.swing.JLabel jLabel1;
    private javax.swing.JLabel jLabel10;
    private javax.swing.JLabel jLabel11;
    private javax.swing.JLabel jLabel2;
    private javax.swing.JLabel jLabel3;
    private javax.swing.JLabel jLabel4;
    private javax.swing.JLabel jLabel5;
    private javax.swing.JLabel jLabel6;
    private javax.swing.JLabel jLabel7;
    private javax.swing.JLabel jLabel8;
    private javax.swing.JLabel jLabel9;
    private javax.swing.JPanel jPanel1;
    private javax.swing.JButton logoutButton;
    private javax.swing.JTextField nameField;
    private javax.swing.JTextField nationalityField;
    private javax.swing.JTextField personalContactNumberField;
    private javax.swing.JButton saveButton;
    // End of variables declaration//GEN-END:variables

```

```
}
```

3. Edit Student Information:

```
package main;
import java.awt.Color;
import java.io.BufferedReader;
import java.io.BufferedWriter;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import static java.lang.System.exit;
import java.util.ArrayList;
import javax.swing.JOptionPane;
/**
 *
 * @author hoors
 */
public class editStudentinfo
{
    public void editName()
    {
        try {
            // Read existing student data from the file
            BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
            ArrayList<String> lines = new ArrayList<>();
            String line;
            while ((line = reader.readLine()) != null) {
                lines.add(line);
            }
            reader.close();

            // Find the line containing the name and update it
            for (int i = 0; i < lines.size(); i++) {
                if (lines.get(i).startsWith("Name:")) {
                    // Prompt the user for the new name
                    String newName = JOptionPane.showInputDialog("Enter the new name:");
                    if (newName == null) {
                        return;
                    }
                    lines.set(i, "Name: " + newName);
                    break; // Once we found and updated the name, we can stop searching
                }
            }

            // Write the modified data back to the file
            BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
            for (String updatedLine : lines) {
                writer.write(updatedLine);
                writer.newLine();
            }
            writer.close();

            JOptionPane.showMessageDialog(null, "Name updated successfully!");
        } catch (IOException e) {
```

```

        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating name.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

public void editdob()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it
        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("Date of Birth:")) {
                // Prompt the user for the new name
                String newDob = JOptionPane.showInputDialog("Enter the new Date of birth:");
                if (newDob == null) {
                    return;
                }
                lines.set(i, "Date of Birth: " + newDob);
                break; // Once we found and updated the name, we can stop searching
            }
        }

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {
            writer.write(updatedLine);
            writer.newLine();
        }
        writer.close();

        JOptionPane.showMessageDialog(null, "Date of Birth updated successfully!");
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating Date of Birth.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

public void editAddress()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
    }
}

```

```

    }
    reader.close();

    // Find the line containing the name and update it
    for (int i = 0; i < lines.size(); i++) {
        if (lines.get(i).startsWith("Address:")) {
            // Prompt the user for the new name
            String newAddress = JOptionPane.showInputDialog("Enter the new Address:");
            if (newAddress == null) {
                return;
            }
            lines.set(i, "Address: " + newAddress);
            break; // Once we found and updated the name, we can stop searching
        }
    }

    // Write the modified data back to the file
    BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
    for (String updatedLine : lines) {
        writer.write(updatedLine);
        writer.newLine();
    }
    writer.close();

    JOptionPane.showMessageDialog(null, "Address updated successfully!");
} catch (IOException e) {
    e.printStackTrace();
    JOptionPane.showMessageDialog(null, "Error occurred while updating Address.", "Error",
JOptionPane.ERROR_MESSAGE);
}
}

public void editEmail()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it
        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("Email:")) {
                // Prompt the user for the new name
                String newEmail = JOptionPane.showInputDialog("Enter the new Email:");
                if (newEmail == null) {
                    return;
                }
                lines.set(i, "Email: " + newEmail);
                break; // Once we found and updated the name, we can stop searching
            }
        }
    }
}

```

```

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {
            writer.write(updatedLine);
            writer.newLine();
        }
        writer.close();

        JOptionPane.showMessageDialog(null, "Email updated successfully!");
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating Email.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

public void editID()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it
        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("ID:")) {
                // Prompt the user for the new name
                String newID = JOptionPane.showInputDialog("Enter the new ID:");
                if (newID == null) {
                    return;
                }
                lines.set(i, "ID: " + newID);
                break; // Once we found and updated the name, we can stop searching
            }
        }

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {
            writer.write(updatedLine);
            writer.newLine();
        }
        writer.close();

        JOptionPane.showMessageDialog(null, "ID updated successfully!");
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating ID.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

```

```

public void editAge()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it
        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("Age:")) {
                // Prompt the user for the new name
                String newAge = JOptionPane.showInputDialog("Enter the new Age:");
                if (newAge == null) {
                    return;
                }
                lines.set(i, "Age: " + newAge);
                break; // Once we found and updated the name, we can stop searching
            }
        }

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {
            writer.write(updatedLine);
            writer.newLine();
        }
        writer.close();

        JOptionPane.showMessageDialog(null, "Age updated successfully!");
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating Age.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

```

```

public void editGender()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it

```

```

        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("Gender: ")) {
                // Prompt the user for the new name
                String newGender = JOptionPane.showInputDialog("Enter the new Gender:");
                if (newGender == null) {
                    return;
                }
                lines.set(i, "Gender: " + newGender);
                break; // Once we found and updated the name, we can stop searching
            }
        }

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {
            writer.write(updatedLine);
            writer.newLine();
        }
        writer.close();

        JOptionPane.showMessageDialog(null, "Gender updated successfully!");
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating Gender.", "Error",
JOptionPane.ERROR_MESSAGE);
    }
}

public void editNationality()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it
        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("Nationality:")) {
                // Prompt the user for the new name
                String newNationality = JOptionPane.showInputDialog("Enter the new Nationality:");
                if (newNationality == null) {
                    return;
                }
                lines.set(i, "Nationality: " + newNationality);
                break; // Once we found and updated the name, we can stop searching
            }
        }

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {

```

```

        writer.write(updatedLine);
        writer.newLine();
    }
    writer.close();

    JOptionPane.showMessageDialog(null, "Nationality updated successfully!");
} catch (IOException e) {
    e.printStackTrace();
    JOptionPane.showMessageDialog(null, "Error occurred while updating Nationality.", "Error",
JOptionPane.ERROR_MESSAGE);
}
}

public void editPersonalContact()
{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it
        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("Personal Contact Number:")) {
                // Prompt the user for the new name
                String newPersonalContactNumber = JOptionPane.showInputDialog("Enter the new Personal Contact
Number:");
                if (newPersonalContactNumber == null) {
                    return;
                }
                lines.set(i, "Personal Contact Number: " + newPersonalContactNumber);
                break; // Once we found and updated the name, we can stop searching
            }
        }

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {
            writer.write(updatedLine);
            writer.newLine();
        }
        writer.close();

        JOptionPane.showMessageDialog(null, "Personal Contact Number updated successfully!");
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating Personal Contact Number.",
"Error", JOptionPane.ERROR_MESSAGE);
    }
}

public void editEmergencyContact()

```



```

{
    try {
        // Read existing student data from the file
        BufferedReader reader = new BufferedReader(new FileReader("Studentdata.txt"));
        ArrayList<String> lines = new ArrayList<>();
        String line;
        while ((line = reader.readLine()) != null) {
            lines.add(line);
        }
        reader.close();

        // Find the line containing the name and update it
        for (int i = 0; i < lines.size(); i++) {
            if (lines.get(i).startsWith("Emergency Contact Number:")) {
                // Prompt the user for the new name
                String newEmergencyContactNumber = JOptionPane.showInputDialog("Enter the new Emergency Contact Number:");
                if (newEmergencyContactNumber == null) {
                    return;
                }
                lines.set(i, "Emergency Contact Number: " + newEmergencyContactNumber);
                break; // Once we found and updated the name, we can stop searching
            }
        }

        // Write the modified data back to the file
        BufferedWriter writer = new BufferedWriter(new FileWriter("Studentdata.txt"));
        for (String updatedLine : lines) {
            writer.write(updatedLine);
            writer.newLine();
        }
        writer.close();

        JOptionPane.showMessageDialog(null, "Emergency Contact Number updated successfully!");
    } catch (IOException e) {
        e.printStackTrace();
        JOptionPane.showMessageDialog(null, "Error occurred while updating Emergency Contact Number.", "Error", JOptionPane.ERROR_MESSAGE);
    }
}

```

```

class editStudentinfoGUI extends javax.swing.JFrame {
    editStudentinfo editinfo;
    public editStudentinfoGUI() {
        editinfo = new editStudentinfo();
        initComponents();
        Color col = new Color(245,210,245);
        getContentPane().setBackground(col);
    }
}

```

```

@SuppressWarnings("unchecked")
// <editor-fold defaultstate="collapsed" desc="Generated Code"> //GEN-BEGIN: initComponents
private void initComponents() {

    jPanel1 = new javax.swing.JPanel();

```

```

jLabel1 = new javax.swing.JLabel();
jButton1 = new javax.swing.JButton();
jButton2 = new javax.swing.JButton();
jButton3 = new javax.swing.JButton();
jButton4 = new javax.swing.JButton();
jButton5 = new javax.swing.JButton();
jButton6 = new javax.swing.JButton();
jButton7 = new javax.swing.JButton();
jButton8 = new javax.swing.JButton();
NameButton = new javax.swing.JButton();
jButton10 = new javax.swing.JButton();
jButton11 = new javax.swing.JButton();
jButton12 = new javax.swing.JButton();
logoutButton = new javax.swing.JButton();

setDefaultCloseOperation(javax.swing.WindowConstants.DISPOSE_ON_CLOSE);
getContentPane().setLayout(new org.netbeans.lib.awtextra.AbsoluteLayout());

jPanel1.setBackground(new java.awt.Color(255, 210, 255));
jPanel1.setLayout(new org.netbeans.lib.awtextra.AbsoluteLayout());

jLabel1.setFont(new java.awt.Font("Segoe Print", 1, 18)); // NOI18N
jLabel1.setText("Select the field you want to edit");
jPanel1.add(jLabel1, new org.netbeans.lib.awtextra.AbsoluteConstraints(150, 30, -1, -1));

jButton1.setBackground(new java.awt.Color(255, 204, 255));
jButton1.setText("Nationality");
jButton1.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton1ActionPerformed(evt);
    }
});
jPanel1.add(jButton1, new org.netbeans.lib.awtextra.AbsoluteConstraints(230, 90, 161, 59));

jButton2.setBackground(new java.awt.Color(255, 204, 255));
jButton2.setText("Personal Contact Number");
jButton2.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton2ActionPerformed(evt);
    }
});
jPanel1.add(jButton2, new org.netbeans.lib.awtextra.AbsoluteConstraints(420, 90, 179, 60));

jButton3.setBackground(new java.awt.Color(255, 204, 255));
jButton3.setText("Address");
jButton3.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton3ActionPerformed(evt);
    }
});
jPanel1.add(jButton3, new org.netbeans.lib.awtextra.AbsoluteConstraints(230, 230, 161, 59));

jButton4.setBackground(new java.awt.Color(255, 204, 255));
jButton4.setText("Date Of Birth");
jButton4.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {

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        jButton4ActionPerformed(evt);
    }
});
jPanel1.add(jButton4, new org.netbeans.lib.awtextra.AbsoluteConstraints(230, 160, 161, 59));

jButton5.setBackground(new java.awt.Color(255, 204, 255));
jButton5.setText("ID");
jButton5.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton5ActionPerformed(evt);
    }
});
jPanel1.add(jButton5, new org.netbeans.lib.awtextra.AbsoluteConstraints(40, 160, 160, 60));

jButton6.setBackground(new java.awt.Color(255, 204, 255));
jButton6.setText("Email");
jButton6.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton6ActionPerformed(evt);
    }
});
jPanel1.add(jButton6, new org.netbeans.lib.awtextra.AbsoluteConstraints(420, 160, 179, 60));

jButton7.setBackground(new java.awt.Color(255, 204, 255));
jButton7.setText("Emergency Contact Number");
jButton7.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton7ActionPerformed(evt);
    }
});
jPanel1.add(jButton7, new org.netbeans.lib.awtextra.AbsoluteConstraints(420, 230, -1, 60));

jButton8.setBackground(new java.awt.Color(255, 204, 255));
jButton8.setText("Gender");
jButton8.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton8ActionPerformed(evt);
    }
});
jPanel1.add(jButton8, new org.netbeans.lib.awtextra.AbsoluteConstraints(40, 300, 160, 60));

NameButton.setBackground(new java.awt.Color(255, 204, 255));
NameButton.setText("Name");
NameButton.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        NameButtonActionPerformed(evt);
    }
});
jPanel1.add(NameButton, new org.netbeans.lib.awtextra.AbsoluteConstraints(40, 90, 160, 59));

jButton10.setBackground(new java.awt.Color(255, 204, 255));
jButton10.setText("Age");
jButton10.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton10ActionPerformed(evt);
    }
});

```

```

});
jPanel1.add(jButton10, new org.netbeans.lib.awtextra.AbsoluteConstraints(40, 230, 160, 60));

jButton11.setBackground(new java.awt.Color(255, 204, 255));
jButton11.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/off.png"))); // NOI18N
jButton11.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton11ActionPerformed(evt);
    }
});
jPanel1.add(jButton11, new org.netbeans.lib.awtextra.AbsoluteConstraints(510, 0, 84, 80));

jButton12.setBackground(new java.awt.Color(255, 204, 255));
jButton12.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/back.png"))); // NOI18N
jButton12.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        jButton12ActionPerformed(evt);
    }
});
jPanel1.add(jButton12, new org.netbeans.lib.awtextra.AbsoluteConstraints(40, 390, 80, 75));

logoutButton.setBackground(new java.awt.Color(255, 204, 255));
logoutButton.setIcon(new javax.swing.ImageIcon(getClass().getResource("/images/logout.png"))); // NOI18N
logoutButton.addActionListener(new java.awt.event.ActionListener() {
    public void actionPerformed(java.awt.event.ActionEvent evt) {
        logoutButtonActionPerformed(evt);
    }
});
jPanel1.add(logoutButton, new org.netbeans.lib.awtextra.AbsoluteConstraints(510, 390, 77, 80));

getContentPane().add(jPanel1, new org.netbeans.lib.awtextra.AbsoluteConstraints(0, 0, 680, 520));

pack();
setLocationRelativeTo(null);
} // </editor-fold> // GEN-END: initComponents

private void jButton7ActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_jButton7ActionPerformed
    editinfo.editEmergencyContact();
} // GEN-LAST:event_jButton7ActionPerformed

private void NameButtonActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_NameButtonActionPerformed
    editinfo.editName();
} // GEN-LAST:event_NameButtonActionPerformed

private void jButton1ActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_jButton1ActionPerformed
    editinfo.editNationality();
} // GEN-LAST:event_jButton1ActionPerformed

private void jButton5ActionPerformed(java.awt.event.ActionEvent evt) { // GEN-
FIRST:event_jButton5ActionPerformed
    editinfo.editID();
} // GEN-LAST:event_jButton5ActionPerformed

```

```

    private void jButton3ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton3ActionPerformed
        editinfo.editAddress();
    }//GEN-LAST:event_jButton3ActionPerformed

    private void jButton10ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton10ActionPerformed
        editinfo.editAge();
    }//GEN-LAST:event_jButton10ActionPerformed

    private void jButton4ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton4ActionPerformed
        editinfo.editedob();
    }//GEN-LAST:event_jButton4ActionPerformed

    private void jButton8ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton8ActionPerformed
        editinfo.editGender();
    }//GEN-LAST:event_jButton8ActionPerformed

    private void jButton2ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton2ActionPerformed
        editinfo.editPersonalContact();
    }//GEN-LAST:event_jButton2ActionPerformed

    private void jButton6ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton6ActionPerformed
        editinfo.editEmail();
    }//GEN-LAST:event_jButton6ActionPerformed

    private void jButton12ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton12ActionPerformed
        this.toBack();
    }//GEN-LAST:event_jButton12ActionPerformed

    private void jButton11ActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_jButton11ActionPerformed
        exit(0);
    }//GEN-LAST:event_jButton11ActionPerformed

    private void logoutButtonActionPerformed(java.awt.event.ActionEvent evt) {//GEN-
FIRST:event_logoutButtonActionPerformed
        this.toBack();
        LoginPage login = new LoginPage();
        login.setVisible(true);
        login.toFront();
    }//GEN-LAST:event_logoutButtonActionPerformed

    // Variables declaration - do not modify//GEN-BEGIN:variables
    private javax.swing.JButton NameButton;
    private javax.swing.JButton jButton1;
    private javax.swing.JButton jButton10;
    private javax.swing.JButton jButton11;
    private javax.swing.JButton jButton12;
    private javax.swing.JButton jButton2;

```

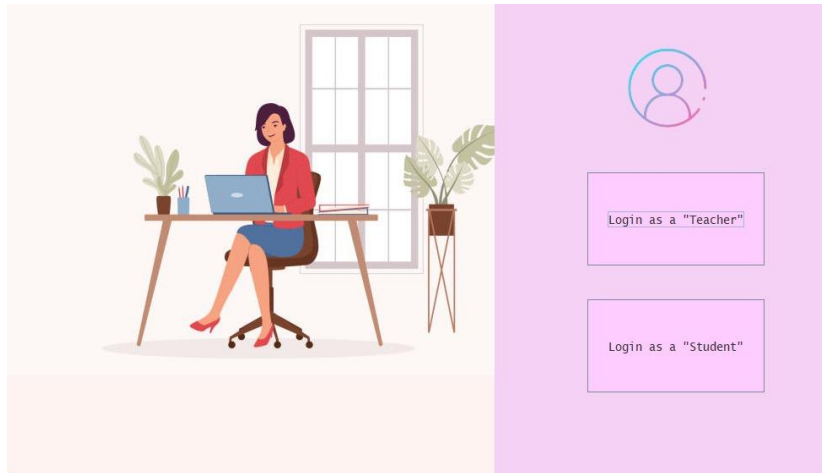
```

private javax.swing.JButton jButton3;
private javax.swing.JButton jButton4;
private javax.swing.JButton jButton5;
private javax.swing.JButton jButton6;
private javax.swing.JButton jButton7;
private javax.swing.JButton jButton8;
private javax.swing.JLabel jLabel1;
private javax.swing.JPanel jPanel1;
private javax.swing.JButton logoutButton;
// End of variables declaration//GEN-END:variables
}

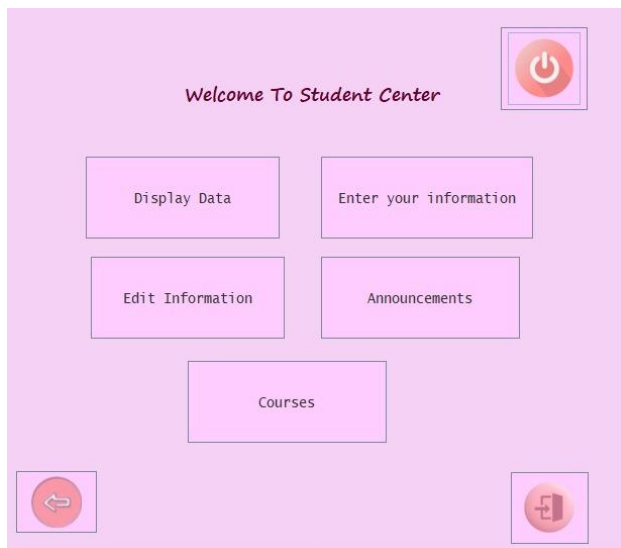
```

Design Phase :

Login Page :



Student Home Page :

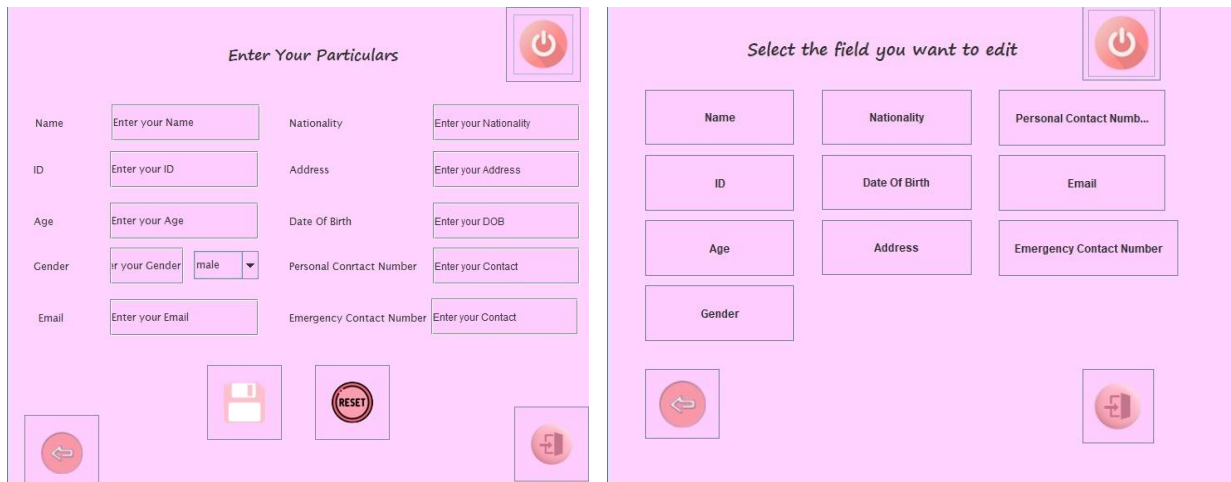


Teacher Home Page :



The Teacher Home Page interface features a light pink background. At the top, it says "Welcome to Teacher Center" next to a red power button icon. Below this, there are four rectangular buttons stacked vertically: "Course", "Announcement", "Student Details", and "Attendance". In the bottom left corner, there is a red circular button with a left-pointing arrow. In the bottom right corner, there is a red circular button with a right-pointing arrow.

CRUD :



The CRUD interface is divided into two panels. The left panel, titled "Enter Your Particulars", contains a form with the following fields: Name (text input), Nationality (text input), ID (text input), Address (text input), Age (text input), Date Of Birth (text input), Gender (dropdown menu with "male" selected), Personal Contact Number (text input), Email (text input), and Emergency Contact Number (text input). At the bottom of this panel are a red circular button with a left-pointing arrow, a red square button with a document icon, a red circular button with the word "RESET", and a red circular button with a right-pointing arrow. The right panel, titled "Select the field you want to edit", contains a grid of buttons for each field: Name, Nationality, Personal Contact Numb..., ID, Date Of Birth, Email, Age, Address, Emergency Contact Number, and Gender. At the bottom of this panel are a red circular button with a left-pointing arrow and a red circular button with a right-pointing arrow.

Division of Responsibilities:

- ❖ **Aqsa Ahmed** was responsible for compiling the document.
- ❖ **Simaab Malik** took charge of providing the interfaces and implemented the sprint.
- ❖ **Hoor Spogmay** took charge of providing the code.